1d toy model of a p-wave superconductor

follow J. Alicea Rep. Prog. Phys. 75, 076501 (2012)

$$H = -\mu \sum_n c_n^+ c_n - \frac{1}{2}\sum_n \left(t\, c_n^+ c_{n+1} + \Delta e^{i\varphi} c_n c_{n+1} + H.c. \right)$$

here $\hat{c}_n$ removes <u>spinless</u> electron from site <u>n</u>.
(Lattice spacing = 1)

$$c_n = \frac{1}{\sqrt{N}} \sum_{k \in BZ} e^{ikn} c_k$$

$$H = -\sum_n \frac{1}{N} \sum_{k_1 k_2} \Big\{ \mu\, c_{k_1}^+ c_{k_2} e^{-in(k_1-k_2)}$$

$$+ \Big[\frac{1}{2} t\, c_{k_1}^+ c_{k_2} e^{-ink_1} e^{i(n+1)k_2} + \frac{\Delta}{2} e^{i\varphi} c_{k_1} c_{k_2} e^{ink_1} e^{i(n+1)k_2}$$

$$+ H.c. \Big] \Big\} = -\sum_{k_1} \Big\{ \mu\, c_{k_1}^+ c_{k_1} + \Big[\frac{1}{2} t\, c_{k_1}^+ c_{k_1} e^{ik_1}$$

(Use $\frac{1}{N}\sum_n e^{in(k_1-k_2)} = \delta_{k_1 k_2}$)

$$+ \frac{\Delta}{2} e^{i\varphi} c_{k_1} c_{-k_1} e^{-ik_1} + H.c. \Big] \Big\}$$

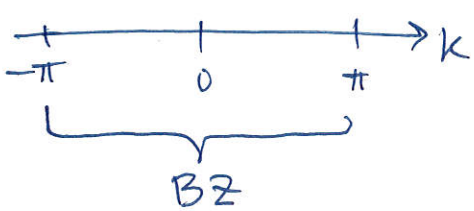
$$\frac{1}{N}\sum_n e^{in(k_1+k_2)} = \delta_{k_1,-k_2}$$

$$H = -\sum_{k \in BZ} \Big\{ (t\cos k + \mu)\, c_k^+ c_k + \frac{1}{2}\Delta e^{i\varphi} e^{-ik} c_k c_{-k}$$

$$+ \frac{\Delta}{2} e^{-i\varphi} e^{ik} c_{-k}^+ c_k^+ \Big\}$$

Check Δ-term: $\sum_k e^{-ik} c_k c_{-k} = \sum_k (\cos k - i\sin k) c_k c_{-k}$

$$\left(\begin{matrix} BZ \\ k \in (-\pi,\pi) \end{matrix} \right) = \sum_{k \in (0,\pi)} (\cos k - i\sin k)\, c_k c_{-k} + \underbrace{\sum_{k \in (-\pi,0)} (\cos k - i\sin k)\, c_k c_{-k}}_{k=-p}$$



$$\sum_{p \in (0,\pi)} (\cos(-p) - i\sin(-p))\, c_{-p} c_p$$

$$= \sum (\cos p + i\sin p)(-c_p c_{-p})$$

Thus $\sum_{k\in BZ} e^{-ik} c_k c_{-k} = \sum_{k\in(0,\pi)} (\cos k - i\sin k)\, c_k c_{-k}$

$$+ \sum_{k\in(0,\pi)} (\cos k + i\sin k)(-c_k c_{-k}) = 2\sum_{k\in(0,\pi)} -i\sin k\, c_k c_{-k}$$

$$= \sum_{k\in(-\pi,\pi)\equiv BZ} (-i\sin k)\, c_k c_{-k}$$ so that only <u>odd function</u> of k is present in this term.

$$H = \sum_{k\in BZ} (-t\cos k - \mu)\, c_k^\dagger c_k - \frac{i}{2}\Delta e^{i\varphi}\sin k\, c_k c_{-k} + \frac{i}{2}\Delta e^{-i\varphi}\sin k\, c_{-k}^\dagger c_k^\dagger$$

$$= \frac{1}{2}\sum_{k\in BZ}\Big\{\varepsilon_k (c_k^\dagger c_k + c_{-k}^\dagger c_{-k}) - i\Delta e^{i\varphi}\sin k\, c_k c_{-k} - i\Delta e^{-i\varphi}\sin k\, c_{-k}^\dagger c_k^\dagger\Big\}$$

$$\varepsilon_k \equiv -t\cos k - \mu, \quad \Delta_k = i\Delta e^{-i\varphi}\sin k \Rightarrow \Delta_k^* = -i\Delta e^{i\varphi}\sin k.$$

Write as a matrix

$$H = \frac{1}{2}\sum_{k\in BZ} (c_k^\dagger, c_{-k}) \begin{pmatrix} \varepsilon_k & -i\Delta e^{-i\varphi}\sin k \\ +i\Delta e^{i\varphi}\sin k & -\varepsilon_k \end{pmatrix} \begin{pmatrix} c_k \\ c_{-k}^\dagger \end{pmatrix} \underbrace{+\varepsilon_k}_{\text{const term.}}$$

Check this: $(c_k^\dagger, c_{-k}) \begin{pmatrix} \varepsilon_k c_k - \Delta_k c_{-k}^\dagger \\ -\Delta_k^* c_k - \varepsilon_k c_{-k}^\dagger \end{pmatrix} =$

$$= \varepsilon_k (c_k^\dagger c_k - c_{-k} c_{-k}^\dagger) - \Delta_k c_k^\dagger c_{-k}^\dagger - \Delta_k^* c_{-k} c_k$$

$$= \varepsilon_k (c_k^\dagger c_k + c_{-k}^\dagger c_{-k}) - \varepsilon_k + \Delta_k c_{-k}^\dagger c_k^\dagger + \Delta_k^* c_k c_{-k}$$

Thus $$H = \frac{1}{2}\sum_k (c_k^+, c_{-k}) \begin{pmatrix} \varepsilon_k & -\Delta_k \\ -\Delta_k^* & -\varepsilon_k \end{pmatrix} \begin{pmatrix} c_k \\ c_{-k}^+ \end{pmatrix}$$

<u>Observe</u>: there is no spin (i.e. all electrons have the same) but the pairing term $\Delta_k = i\Delta e^{-i\varphi} \sin k$ is <u>odd function</u> of k $\Rightarrow$ is of p-wave symmetry.

To diagonalize, we try

$$\begin{pmatrix} c_k \\ c_{-k}^+ \end{pmatrix} = \begin{pmatrix} \cos\theta_k & \sin\theta_k e^{i\beta_k} \\ -\sin\theta_k e^{-i\beta_k} & \cos\theta_k \end{pmatrix} \begin{pmatrix} d_k \\ d_{-k}^+ \end{pmatrix}$$

where θ and β are functions of $\underline{k}$ to be determined.

From $c_k = \cos\theta_k d_k + \sin\theta_k e^{i\beta_k} d_{-k}^+$,

obtain $c_k^+ = \cos\theta_k d_k^+ + \sin\theta_k e^{-i\beta_k} d_{-k}$,

compare with 2nd line: $c_{-k}^+ = \cos\theta_k d_{-k}^+ - \sin\theta_k e^{-i\beta_k} d_k$

to deduce that θ_k is <u>odd function of k</u>

β_k is even.

Transpose and conjugate:

$$(c_k^+, c_{-k}) = \begin{pmatrix} c_k \\ c_{-k}^+ \end{pmatrix}^+ = (d_k^+, d_{-k}) \begin{pmatrix} \cos\theta_k & -\sin\theta_k e^{i\beta} \\ \sin\theta_k e^{-i\beta} & \cos\theta_k \end{pmatrix}$$

Plug in H and simplify

$$H = \frac{1}{2}\sum_k (d_k^+, d_{-k}) \underbrace{\begin{pmatrix} \cos\theta & -\sin\theta e^{i\beta} \\ \sin\theta e^{-i\beta} & \cos\theta \end{pmatrix}\begin{pmatrix} \varepsilon & -\Delta \\ -\Delta^* & -\varepsilon \end{pmatrix}\begin{pmatrix} \cos\theta & \sin\theta e^{i\beta} \\ -\sin\theta e^{-i\beta} & \cos\theta \end{pmatrix}}_{\text{all at the same } k.} \begin{pmatrix} d_k \\ d_{-k}^+ \end{pmatrix}$$

$$A = \begin{pmatrix} \cos\theta & -\sin\theta e^{i\beta} \\ \sin\theta e^{-i\beta} & \cos\theta \end{pmatrix}\begin{pmatrix} \varepsilon\cos\theta + \Delta\sin\theta e^{-i\beta} & \varepsilon\sin\theta e^{i\beta} - \Delta\cos\theta \\ -\Delta^*\cos\theta + \varepsilon\sin\theta e^{-i\beta} & -\Delta^*\sin\theta e^{i\beta} - \varepsilon\cos\theta \end{pmatrix} =$$

$$= \begin{pmatrix} \varepsilon\cos^2\theta + \Delta\cos\theta\sin\theta e^{-i\beta} + \Delta^*\sin\theta\cos\theta e^{i\beta} - \varepsilon\sin^2\theta; & 2\varepsilon\cos\theta\sin\theta e^{i\beta} - \Delta\cos^2\theta + \Delta^*\sin^2\theta e^{i2\beta} \\ 2\varepsilon\cos\theta\sin\theta e^{-i\beta} + \Delta\sin^2\theta e^{-i2\beta} - \Delta^*\cos^2\theta; & \varepsilon\sin^2\theta - \Delta\sin\theta\cos\theta e^{-i\beta} - \Delta^*\cos\theta\sin\theta e^{i\beta} - \varepsilon\cos^2\theta \end{pmatrix}$$

Observe that diagonal elements ~~have~~ are opposite to each other.

<u>Require</u> off-diagonal elements to be <u>zero</u>:

upper right: $\varepsilon\sin(2\theta)e^{i\beta} = \Delta\cos^2\theta - \Delta^*\sin^2\theta\, e^{i2\beta}$

$\varepsilon\sin(2\theta) = \Delta e^{-i\beta}\cos^2\theta - \Delta^* e^{i\beta}\sin^2\theta$

lower left: $\varepsilon\sin(2\theta)e^{-i\beta} = \Delta^*\cos^2\theta - \Delta\sin^2\theta\, e^{-i2\beta}$

$\varepsilon\sin(2\theta) = \Delta^* e^{i\beta}\cos^2\theta - \Delta\sin^2\theta\, e^{-i\beta}$

Choose phase $e^{i\beta}$ such that: $\begin{cases} \Delta_k e^{-i\beta} = \text{real} \\ \Delta_k^* e^{i\beta} = \text{real} \end{cases}$

$(i\Delta e^{-i\varphi}\sin k)e^{-i\beta} = \text{real}$

$i e^{-i(\varphi+\beta)} = 1 \Rightarrow \beta = -\varphi + \frac{\pi}{2} \Rightarrow i e^{-i(\varphi-\varphi+\frac{\pi}{2})} = i(-i) = 1$

then $\Delta_k^* e^{i\beta} = (-i\Delta e^{i\varphi}\sin k)\, e^{i(-\varphi+\frac{\pi}{2})} = \underbrace{-i(-i)}_{+1}\Delta\sin k$

works!

Thus we have one equation to satisfy now:

$$\varepsilon_k \sin(2\theta_k) = \Delta \sin k \cos^2\theta_k - \Delta \sin k \sin^2\theta_k =$$

$$= (\Delta \sin k) \cos(2\theta_k)$$

$$\tan(2\theta_k) = \frac{\Delta \sin k}{\varepsilon_k} \equiv \frac{f_k}{\varepsilon_k}$$

Here $f_k \equiv \Delta \sin k$ is real function of k by construction, and it is <u>odd</u> as expected.

Then $A = \begin{pmatrix} \varepsilon(\cos^2\theta - \sin^2\theta) + \Delta \sin k \cdot \sin 2\theta; & 0 \\ 0; & -\varepsilon \cos 2\theta - \Delta \sin k \sin 2\theta \end{pmatrix}$ takes on diagonal form.

$$E_k = \varepsilon_k \cos 2\theta_k + f_k \sin 2\theta_k = \cos 2\theta_k \left(\varepsilon_k + f_k \tan(2\theta_k)\right) =$$

$$= \frac{1}{\sqrt{1+\tan^2(2\theta_k)}} \left(\varepsilon_k + f_k \cdot \frac{f_k}{\varepsilon_k}\right) = \frac{\varepsilon_k^2 + f_k^2}{\varepsilon_k} \frac{\varepsilon_k}{\sqrt{\varepsilon_k^2 + f_k^2}} =$$

$$= \sqrt{\varepsilon_k^2 + f_k^2} = \sqrt{\varepsilon_k^2 + |\Delta_k|^2} = \sqrt{(t\cos k + \mu)^2 + \Delta^2 \sin^2 k}$$

$$\Rightarrow H = \frac{1}{2} \sum_k (d_k^+, d_{-k}) \begin{pmatrix} E_k & 0 \\ 0 & -E_k \end{pmatrix} \begin{pmatrix} d_k \\ d_{-k}^+ \end{pmatrix} =$$

$$= \frac{1}{2} \sum_k E_k \left(d_k^+ d_k - d_{-k} d_{-k}^+\right) = \sum_k E_k d_k^+ d_k .$$

$$\cos(2\theta_k) = \frac{\varepsilon_k}{\sqrt{\varepsilon_k^2 + f_k^2}} = \frac{\varepsilon_k}{E_k}$$

$$\parallel$$

$$2\cos^2(\theta_k) - 1 \quad \Rightarrow \quad \cos^2\theta_k = \frac{1}{2}\left(1 + \frac{\varepsilon_k}{E_k}\right)$$

$$\cos\theta_k = \sqrt{\frac{E_k + \varepsilon_k}{2E_k}}$$

$$\sin(2\theta_k) = \frac{f_k}{\varepsilon_k}\cos(2\theta_k) = \frac{f_k}{E_k}$$

$$\parallel$$

$$2\sin\theta_k\cos\theta_k \quad \Rightarrow \quad \sin\theta_k = \frac{f_k}{2E_k}\frac{1}{\cos\theta_k} = \frac{f_k}{2E_k}\sqrt{\frac{2E_k}{E_k+\varepsilon_k}} =$$

$$= \frac{f_k}{\sqrt{2E_k}}\sqrt{\frac{E_k - \varepsilon_k}{(E_k+\varepsilon_k)(E_k-\varepsilon_k)}} = \frac{f_k}{|f_k|}\sqrt{\frac{E_k-\varepsilon_k}{2E_k}}$$

$$E_k^2 - \varepsilon_k^2 = f_k^2$$

Important: since $f_k = \Delta \sin K$ is odd,

$$\frac{f_k}{|f_k|} = \frac{\sin K}{|\sin K|} = \text{sign}(\sin(K)) \text{ is odd too.}$$

And $\tan(\theta_k) = \dfrac{\sin\theta_k}{\cos\theta_k} = \dfrac{f_k}{|f_k|}\sqrt{\dfrac{E_k-\varepsilon_k}{E_k+\varepsilon_k}} =$

$$= \frac{f_k}{|f_k|}\frac{\sqrt{E_k^2 - \varepsilon_k^2}}{E_k+\varepsilon_k} = \frac{f_k}{E_k+\varepsilon_k}$$

OR, $= \dfrac{f_k}{|f_k|}\dfrac{E_k-\varepsilon_k}{\sqrt{E_k^2-\varepsilon_k^2}} = \dfrac{f_k}{f_k^2}(E_k-\varepsilon_k) = \dfrac{E_k-\varepsilon_k}{f_k}$

Let us construct the ground state of

$$H = \sum_{k \in BZ} E_k \, d_k^\dagger d_k$$ $\Rightarrow$ clearly it must be vacuum of d-particles, i.e. such a state $|g\rangle$ that

$$d_k |g\rangle = 0 \quad \text{for any } k \in BZ.$$

But such a state is constructed from

$$|g\rangle = \prod_{\substack{k \in BZ \\ [k \in (-\pi, \pi)]}} d_k |0\rangle$$ where $|0\rangle$ is the ground state of <u>c-particles</u>! (state with no original fermions)

By construction, $d_p |g\rangle = d_p \prod_k d_k |0\rangle =$

$$= \prod_{k \neq p} d_k (d_p)^2 |0\rangle = 0$$

because $(d_p)^2 = 0$ always.

Let us now re-group d_k's in $|g\rangle$ and write

$$|g\rangle = \prod_{k \in (0, \pi)} d_{-k} d_k |0\rangle.$$

p. 181

Invert $\begin{pmatrix} c_k \\ c_{-k}^\dagger \end{pmatrix} = \begin{pmatrix} \cos\theta_k & \sin\theta_k e^{i\beta_k} \\ -\sin\theta_k e^{-i\beta_k} & \cos\theta_k \end{pmatrix} \begin{pmatrix} d_k \\ d_{-k}^\dagger \end{pmatrix}$

to get $\begin{pmatrix} d_k \\ d_{-k}^\dagger \end{pmatrix} = \begin{pmatrix} \cos\theta_k & -\sin\theta_k e^{i\beta_k} \\ \sin\theta_k e^{-i\beta} & \cos\theta_k \end{pmatrix} \begin{pmatrix} c_k \\ c_{-k}^\dagger \end{pmatrix}$

Thus $d_{-k} = \cos\theta_{-k}\, c_{-k} - \sin\theta_{-k}\, e^{i\beta_{-k}}\, c_k^+ =$

$= (\sin\theta_k \text{ is odd!}) = \cos\theta_k\, c_{-k} + \sin\theta_k\, e^{i\beta_k}\, c_k^+$

then $d_{-k}\, d_k |0\rangle = \left(\cos\theta_k\, c_{-k} + \sin\theta_k\, e^{i\beta_k}\, c_k^+\right)\left(\cos\theta_k\, c_k - \sin\theta_k\, e^{i\beta}\, c_{-k}^+\right)|0\rangle =$

$= \Big(\cos^2\theta_k\, c_{-k} c_k - \cos\theta_k \sin\theta_k\, e^{i\beta}\, c_{-k} c_{-k}^+$

$+ \cos\theta_k \sin\theta_k\, e^{i\beta}\, c_k^+ c_k - \sin^2\theta_k\, e^{i2\beta}\, c_k^+ c_{-k}^+\Big)|0\rangle$

$= \left(c_k|0\rangle = 0 \text{ for any } k \text{ and } c_{-k} c_{-k}^+ = 1 - c_{-k}^+ c_{-k}\right) =$

$= \left(-\cos\theta_k \sin\theta_k\, e^{i\beta} - \sin^2\theta_k\, e^{i2\beta}\, c_k^+ c_{-k}^+\right)|0\rangle$

$= \left(-\cos\theta_k \sin\theta_k\, e^{i\beta}\right)\left(1 + \frac{\sin\theta_k}{\cos\theta_k}\, e^{i\beta}\, c_k^+ c_{-k}^+\right)|0\rangle$

From p. 181, 184: $\tan\theta_k\, e^{i\beta} = \dfrac{E_k - \varepsilon_k}{f_k e^{-i\beta}} = \dfrac{E_k - \varepsilon_k}{\Delta \sin k\, e^{i\varphi} e^{-i\pi/2}} = \dfrac{E_k - \varepsilon_k}{-i\Delta e^{i\varphi} \sin k} =$

$$= \frac{E_k - \varepsilon_k}{\Delta_k^*}$$

Thus we obtain <u>BCS</u> wave-function for p-wave SC

$$|g\rangle \sim \prod_{k\in(0,\pi)} \left(1 + \frac{E_k - \varepsilon_k}{\Delta_k^*}\, c_k^+ c_{-k}^+\right)|0\rangle$$

Now back to c-form of H at p. 180:

$$H = \sum_{k \in BZ} (c_k^+, c_{-k})\, \mathcal{H}_k \begin{pmatrix} c_k \\ c_{-k}^+ \end{pmatrix}$$

$$\mathcal{H}_k = \frac{1}{2}\begin{pmatrix} \varepsilon_k & -i\Delta e^{-i\varphi}\sin k \\ i\Delta e^{i\varphi}\sin k & -\varepsilon_k \end{pmatrix} = \vec{h}(k)\cdot\vec{\sigma} = h_x\sigma^x + h_y\sigma^y + h_z\sigma^z$$

$$= \begin{pmatrix} h_z & h_x - i h_y \\ h_x + i h_y & -h_z \end{pmatrix}$$

Thus $\begin{cases} h_x - i h_y = -i\Delta e^{-i\varphi}\sin k \\ h_z = \varepsilon_k \end{cases}$

Hence vector $\vec{h}(k)$ has the following features:

$$h_z(k) = h_z(-k)$$

but $h_{x,y}(k) = -h_{x,y}(-k)$

Thus the map from $k \in BZ = [-\pi, \pi]$ to the unit sphere $\hat{n}(k) = \dfrac{\vec{h}(k)}{|\vec{h}(k)|}$ is fully specified by $k \in [0, \pi]$ range.

Oddness of $h_{x,y}(k)$ means that $h_{x,y}(0) = 0$

$$h_{x,y}(-\pi) = h_{x,y}(\pi) = 0$$

$\parallel$

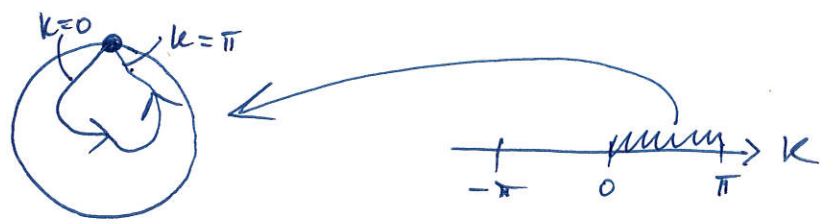
$-h_{x,y}(\pi)$ by oddness

because $-\pi = \pi + \underbrace{2\pi}_{\text{reciprocical lattice vector.}}$

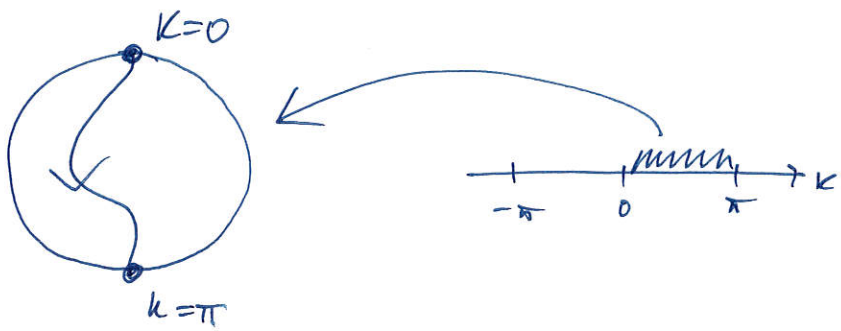
Thus at $k=0$ and $k=\pm\pi$ $\hat{n}(k) = \text{sign}(h_z(k))$
i.e. it can only point to North/South poles.

Hence we see the emergence of two distinct topological classes:

(A) $\hat{n}(k=0) = \hat{n}(k=\pi)$ $\left(= \hat{n}(k=-\pi) \text{ of course}\right)$.



(B) $\hat{n}(k=0) = -\hat{n}(k=\pi)$



Now, $h_z(k) = -t\cos k - \mu = \begin{cases} -t-\mu, & k=0 \\ t-\mu, & k=\pi(\pm) \end{cases}$

Take $\underline{\underline{t>0}}$: consider $\mu<0$, $\mu=-q$ $(q>0)$

$$\hat{n}(0) = -\text{sign}(t+\mu) = -\text{sgn}(t-q) = \text{sgn}(q-t)$$

$$\hat{n}(\pi) = \text{sgn}(t-\mu) = \text{sgn}(t+q)$$

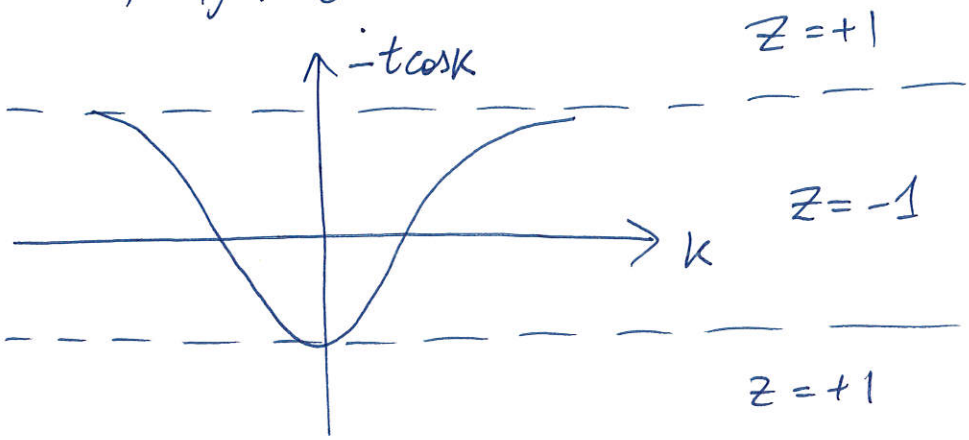
$\hat{n}(0) = \hat{n}(\pi)$ if $q>t$

$\hat{n}(0) = -\hat{n}(\pi)$ if $q<t$ $\Rightarrow$ change at $q=t$, $\mu=-t$

Most convenient is to consider product:

$$Z = \hat{n}(0)\cdot\hat{n}(\pi) = -\mathrm{sign}(t+\mu)\,\mathrm{sign}(t-\mu) = \mathrm{sign}(\mu^2 - t^2)$$

$$\left.\begin{array}{l} \text{If } Z=+1, \ |\mu|>t \\ \text{If } Z=-1, \ |\mu|<t \end{array}\right\} Z_2\text{-invariant}$$



Majorana formulation

let $\hat{c}_n = \frac{1}{2} e^{-i\varphi/2} \left(\hat{\gamma}_{B,n} + i\hat{\gamma}_{A,n}\right)$

where $\hat{\gamma}_{\alpha,n} = \hat{\gamma}^{+}_{\alpha,n}$ is Majorana fermion $(\alpha = A, B)$

Commutation relation $\{\hat{\gamma}_{\alpha,n}, \hat{\gamma}_{\alpha',n'}\} = 2\,\delta_{\alpha\alpha'}\,\delta_{nn'}$

Note $\hat{c}^{+}_n = \frac{1}{2} e^{i\varphi/2}\left(\hat{\gamma}^{+}_{B,n} - i\hat{\gamma}^{+}_{A,m}\right) = \frac{1}{2} e^{i\varphi/2}\left(\hat{\gamma}_{B,n} - i\hat{\gamma}_{A,n}\right)$

Then $\sum_n \hat{c}^{+}_n \hat{c}_n = \sum_n \frac{1}{4}\left(\gamma_{B,n} - i\gamma_{A,n}\right)\left(\gamma_{B,n} + i\gamma_{A,n}\right) =$

$$= \frac{1}{4}\sum_n \left(\gamma^2_{B,n} + \gamma^2_{A,m} + i\,\gamma_{B,n}\gamma_{A,n} - i\,\gamma_{A,n}\gamma_{B,n}\right)$$

But $\{\gamma_{B,n}, \gamma_{B,n}\} = 2\gamma^2_{B,n} = 2 \Rightarrow \gamma^2_{B,n} = 1.$

$$\{\gamma_{B,n}, \gamma_{A,n}\} = 0 \Rightarrow \gamma_{B,n}\gamma_{A,n} = -\gamma_{A,n}\gamma_{B,n}$$

$$\Rightarrow \sum_n c^{+}_n c_n = \frac{1}{2}\sum_n \left(1 + i\,\gamma_{B,n}\gamma_{A,n}\right)$$

$$\sum_n \Delta e^{i\varphi} c_n c_{n+1} = \frac{1}{4}\Delta \sum_n \left(\gamma_{B,n} + i\gamma_{A,n}\right)\left(\gamma_{B,n+1} + i\,\gamma_{A,n+1}\right) =$$

$$= \frac{1}{4}\Delta \sum_n \left(\gamma_{B,n}\gamma_{B,n+1} - \gamma_{A,n}\gamma_{A,n+1} + i\,\gamma_{B,n}\gamma_{A,n+1} + i\,\gamma_{A,n}\gamma_{B,n+1}\right)$$

But H.c. of that is $\sum_n \Delta e^{-i\varphi} c^{+}_{n+1} c^{+}_n = \frac{1}{4}\Delta \sum_n \left(\gamma_{B,n+1} - i\gamma_{A,n+1}\right)\cdot$

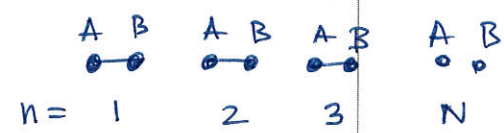
$$\cdot\left(\gamma_{B,n} - i\gamma_{A,n}\right) = \frac{1}{4}\Delta \sum_n \left(\gamma_{B,n+1}\gamma_{B,n} + \gamma_{A,n+1}\gamma_{A,n} - i\,\gamma_{A,n+1}\gamma_{B,n} - i\,\gamma_{B,n+1}\gamma_{A,n}\right)$$

At the end of the day

$$H = -\frac{\mu}{2}\sum_{n=1}^{N}\left(1+i\gamma_{B,n}\gamma_{A,n}\right) - \frac{1}{4}\sum_{n=1}^{N-1}\Big\{(\Delta+t)\,\gamma_{B,n}\gamma_{A,n+1} + (\Delta-t)\,\gamma_{A,n}\gamma_{B,n+1}\Big\}$$

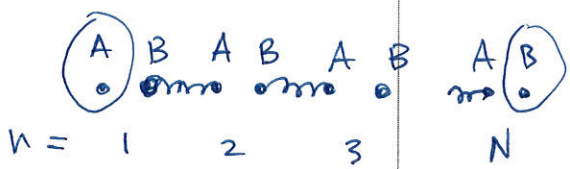
Two simple but important limits:

(A) $\Delta = t = 0$ but $\mu < 0$



$$\Rightarrow H_A = \frac{|\mu|}{2}\sum_{n=1}^{N}\left(1+i\gamma_{B,n}\gamma_{A,n}\right)$$

(B) $\mu = 0$ and $t = \Delta \neq 0$



$$H_B = -i\frac{t}{2}\sum_{n=1}^{N-1}\gamma_{B,n}\gamma_{A,n+1}$$

let $f_n = \frac{1}{2}\left(\gamma_{A,n+1} + i\gamma_{B,n}\right)$

$$f_n^+ = \frac{1}{2}\left(\gamma_{A,n+1} - i\gamma_{B,n}\right)$$

$$\gamma_{A,n+1} = f_n + f_n^+$$

$$\gamma_{B,n} = -i\left(f_n - f_n^+\right)$$

Note: $\gamma_{A,1}$ and $\gamma_{B,N}$ do not appear in H_B.
$\Rightarrow$ form non-local $F = \frac{1}{2}\left(\gamma_{A,1} + i\gamma_{B,N}\right)$ zero-energy mode

$$\Rightarrow H_B = -\frac{it}{2}\sum_{n=1}^{N-1}(-i)\left(f_n - f_n^+\right)\left(f_n + f_n^+\right) =$$

$$= -\frac{t}{2}\sum_{n=1}^{N-1}\Big\{\underbrace{f_n^2}_{=0} - \underbrace{\left(f_n^+\right)^2}_{=0} + \underbrace{f_n f_n^+}_{1-f_n^+ f_n} - f_n^+ f_n\Big\} = t\sum_{n=1}^{N-1}\left(f_n^+ f_n - \frac{1}{2}\right) + 0\cdot\left(F^+F\right)$$

Hence ground state of H_B is two-fold degenerate:

states $|0\rangle$ (such that $F|0\rangle = 0$)

and $|1\rangle$ ($|1\rangle = F^+|0\rangle$ so that $F|1\rangle = FF^+|0\rangle =$

$= (1 - F^+F)|0\rangle = |0\rangle$)

are degenerate in energy

but different $\langle 0|1\rangle = \langle 0|F^+|0\rangle = 0$.

$|0\rangle$ and $|1\rangle$ describe empty (filled) zero-energy Majorana end-states which are highly <u>non-local</u>

Note: for $\Delta \neq t$ and $\mu \neq 0$ Majorana end-states persist although are less localized (decay from boundaries as $e^{-x/\xi}$)

and are split in energy $\varepsilon_0 - \varepsilon_1 \sim e^{-L/\xi}$

where L = length of the chain.